

IBRAHIM RATHWALA

Dubai, UAE | +971 56 109 3935 | ibrahimrathwala@gmail.com | [Portfolio](#) | linkedin.com/in/ibrahim-rathwala
UAE Golden Residency Visa (no sponsorship required) | UAE Driving License | English, Arabic, Urdu, Gujarati, Hindi

SUMMARY

AI/ML Engineer with First-Class Honours in Computer Science (Artificial Intelligence) and hands-on experience building agentic AI systems, LLM-powered workflow automation, and full-stack data platforms. Currently pursuing an MS in Computer Science (Machine Learning) at Georgia Tech. Ship production AI systems serving 20 companies and platforms that cut operational costs by 59%.

SKILLS

Languages: Python, SQL, Java, R, C, JavaScript

ML / AI: LLMs & Agentic AI (GPT-4o, LangChain), RAG, FAISS, sentence-transformers, NLP, PyTorch, TensorFlow, Scikit-learn

Data: Pandas, NumPy, PostgreSQL, MySQL, Redis, Data Cleaning & ETL

Tools / Cloud: FastAPI, Next.js, React, Docker, GitHub Actions (CI/CD), AWS S3, Microsoft Azure, Git

EXPERIENCE

AI/ML Engineer — The Rufescent

Dubai, UAE | June 2025 – Present

- Built agent modules for HÜSE.AI, a multi-tenant conversational HR platform on WhatsApp — leave application and approval, attendance and regularisation, holidays, meeting scheduling and mood check-ins — serving 20 companies under a single group hierarchy.
- Engineered the leave balance engine and policy RAG — FAISS + sentence-transformers retrieval over policy documents via LangChain, with GPT-4o / GPT-4o-mini tiered per task to control per-message cost.
- Led an audit of the attendance and reporting stack that found 25 correctness defects across nine conflicting status ladders — the same employee-day reading "Present" on web, "Week Off" via API and "Not rostered" by email — and shipped fixes for 24, verified live against 140 employees.
- Delivered the reporting and session-security layer of the SOP Orchestrator (FastAPI, React/TypeScript): consolidated attendance and productivity dashboards, bulk leave approvals, punch-session and geo-location visibility, and cross-device session revocation.
- Built the Integration API v1 connecting HÜSE Core to the Orchestrator — 14 endpoints across 7 domains (companies, groups, shifts, rotations, leave requests, attendance) with scoped auth, rate limiting and audit logging.

Software Engineer Intern — UKR Group

Dubai, UAE | June 2024 – September 2024

- Developed a custom CRM (PHP, MySQL) with Python-based data-cleaning pipelines, streamlining client data integration and improving data accuracy.
- Implemented an eSignature platform for contract signing and email dispatch, cutting operational costs by 59%.

EDUCATION

Georgia Institute of Technology — MS, Computer Science (Machine Learning)

Online | 2026 – Present

Heriot-Watt University — BSc Computer Science (AI), First-Class Honours

Dubai, UAE | 2022 – 2025

PROJECTS

Admin Operations Management System (AOMS) [\[live demo\]](#)

- Built a multi-tenant operations platform (FastAPI, PostgreSQL, Redis, React, React Native) with 320+ REST endpoints spanning an admin dashboard, field mobile app, and system-admin console.
- Engineered a double-entry accounting engine posting every payment, expense, invoice, and asset event as immutable balanced journal entries — powering P&L, balance sheet, cash-flow, and AR/AP aging reports with PDF export.
- Secured with JWT + TOTP 2FA, Redis session revocation, role-based permissions, and an append-only audit trail; deployed via Docker Compose with GitHub Actions CI.

Venigma — AI Video Generation & Captioning Tool [\[GitHub\]](#)

- Developed a web app (Next.js, Flask) for real-time text-to-video generation from user prompts using Hugging Face models and Llama 3 (Groq), with an automatic captioning pipeline — audio extraction + Azure Cognitive Services Speech SDK — producing accurate, time-aligned captions for uploaded videos.

AWARDS & CERTIFICATIONS

- Deputy Principal's Award for Academic Excellence (2023–2025), Heriot-Watt University — three consecutive years.
- RTA × Alibaba Hackathon — Best Solution Champion (3rd place overall) — UAE taxi-services solution.
- Microsoft Certified: Azure AI Fundamentals (AI-900) · Coursera: Machine Learning Specialization